Ai-Plan: Learning Simulator-Corrected Markov Kernels for Action-Chunk Refinement

Concept manuscript I: a fixed scene and a task-defined target

Concept manuscript. Methods and proofs are proposed derivations. All experimental findings and numerical results are explicitly hypothetical; no experiments reported here were run.

Abstract

A robot that revises its previous action chunk must correct plans at states caused by its own execution, where a recorded expert future is no longer a valid label. We study the setting in which a simulator and a task cost are available, but an expert cannot be queried. We freeze a behavior-cloned stochastic chunk generator and define a planning distribution by exponentially tilting its output law with simulated cost. We show that this distribution can be sampled in the generator's noise space without evaluating its action density. Inspired by Ai-Sampler, we learn proposal and acceptance models through a one-step probability-balance objective. We derive a bound from acceptance error to stationary error under an explicit exact-refresh mechanism, and distinguish distributional correctness from finite-budget mixing. The resulting sampler can serve as an online planner or produce correction targets for an amortized revision policy. A clearly marked hypothetical experimental section illustrates evaluation on ambiguous-route repair and contact-rich pushing. The results there are not measurements.

1. Introduction

How can a generative robot policy improve a stale plan when no expert can tell it what to do next? Behavior cloning supplies a repertoire of plausible actions, but it does not label recovery states reached after an imperfect push or an unexpected object displacement. A simulator makes those states observable and allows alternative commands to be executed in copies of the environment. It still does not make the original demonstration a correct continuation from the new state.

We consider a different source of supervision: a task cost evaluated through simulation. The objective is not to discard the behavior-cloned generator and solve control from scratch. Instead, we seek corrections that remain plausible under that generator while assigning greater probability to plans with favorable simulated consequences. This yields a conditional distribution over good plans rather than a single optimized trajectory.

Our starting point is the fixed-point perspective of Egorov et al. (2024): learn a Markov kernel whose application preserves a desired distribution, and use a balance discrepancy to train the kernel. We transfer this idea to robot planning at a *fixed scene*. The target is no longer an unknown expert conditional density. It is a specified simulator-cost tilt of a pretrained action-chunk prior. A useful construction is to place this target on the generator's random inputs, where the base density is known even when the action likelihood is intractable.

This construction separates two questions that are often conflated. First, does repeated refinement have the correct invariant distribution? Second, does a short refinement run find useful alternatives? Exact Metropolis–Hastings correction answers the first but not the second. A learned approximate correction can reduce simulator calls, but then its stationary bias must be controlled. A nearly identity kernel illustrates why a small one-step invariance loss is not sufficient without a mixing condition.

We develop three ingredients. We first derive the simulator-defined target and its noise-space representation. We then construct a learned acceptance model with swap antisymmetry and derive a probability-flow residual that includes both forward and reverse acceptance factors. Finally, we add an occasional exact global refresh, obtaining an explicit finite-step error bound. The bridge decoder and its dissipative prior determine which plans are easy to propose; they do not supply Metropolis–Hastings correctness. Our hypothetical experiments are designed to separate these roles.

2. Related Work

Learning invariant kernels. Hastings (1970) establishes the acceptance correction underlying our exact sampler. Implicit Metropolis–Hastings (Neklyudov et al., 2019) relates local invariance discrepancies to stationary bias through a minorization condition. Ai-Sampler (Egorov et al., 2024) combines learned involutive proposals, symmetry-aware discrimination, and MH bootstrapping. We inherit its fixed-

point viewpoint, but rederive the balance objective for an explicit stochastic proposal in the noise space of a robot policy. In particular, our approximate-kernel theorem states its mixing assumption rather than identifying small one-step error with stationary accuracy unconditionally.

Generative policies and bridges. Diffusion Policy (Chi et al., 2023) learns conditional action sequences. Flow matching (Lipman et al., 2023), conditional diffusion bridges (Shi et al., 2022), and diffusion Schrödinger bridge matching (DSBM; Shi et al., 2023) provide alternative generators. We use a conditional DSBM decoder as a prior, with a physics-informed revision reference where tractable; Tang (2026) supplies a tutorial treatment. Our sampling construction also applies to FM and diffusion decoders. Neither non-Gaussian initialization nor an MH correction is exclusive to SBs.

Sampling-based planning. Information-theoretic MPC (Williams et al., 2018) uses cost-dependent changes of measure, while Stein variational MPC (Lambert et al., 2021) maintains interacting control candidates. Our planning target is closely related, but the proposal space is the random-input space of a cached-plan reviser. A particularly strong baseline is preconditioned Crank–Nicolson (pCN; Cotter et al., 2013), which preserves a Gaussian noise prior and requires only a simulator-cost ratio for acceptance. Learning is valuable only if it improves finite-budget exploration over such simple exact samplers.

3. Preliminaries and Problem Setting

An action chunk is $A = (a_0, \dots, a_{H-1}) \in \mathbb{R}^{Hd_a}$, with H commands and command dimension d_a. At each robot replan, only the first $K \le H$ commands are executed. The remaining suffix is aligned and completed to form the next cached proposal C. The completion rule is fixed during the analysis.

We write $x = (s, h, C, g)$ for a planning context: s is the full simulator snapshot, h is the available observation/action history, and g is the task goal. The same snapshot is restored before evaluating every candidate. The learned networks may use only the observable parts of x; full simulator state is needed by the evaluator, not necessarily exposed to the policy.

Let B_x be a pretrained stochastic process that revises a whole chunk over artificial time $\tau \in [0, 1]$, starting from $Y_0 = C$ and ending at $A = Y_1$. We call its terminal law b_x. Revision time is not physical execution time. A finite-step implementation is a deterministic map of random innovations:

$$A = G_\psi(x, z), \qquad z \sim \varphi = \mathcal{N}(0, I_d), \tag{1}$$

where d is the number of scalar random inputs and ψ contains the frozen decoder parameters. The notation $G_{\psi\#}\varphi = b_x$ denotes the pushforward distribution. The map need not be invertible.

A cost $J_x(A)$ is computed by executing a candidate prefix in a simulator copy and, if needed, continuing with a fixed rollout policy. This continuation carries the candidate's suffix, so two chunks with identical executed prefixes can still receive different long-horizon evaluations. We initially assume deterministic simulation and $0 \le J_x(A) \le J_{\max} < \infty$. A bounded cost is an explicit modeling assumption used for the quantitative refresh bound; it is not a property of arbitrary robotics objectives.

For probability laws p, q, we use $\mathrm{TV}(p, q) = \sup_E |p(E) - q(E)|$, where E ranges over measurable events. The relative entropy is $\mathrm{KL}(p\|q) = \int \log(\mathrm{d}p/\mathrm{d}q)\,\mathrm{d}p$ when p is absolutely continuous with respect to q, and infinity otherwise.

4. Method

4.1. A planning distribution without expert labels

For a temperature $\beta > 0$, define

$$\begin{aligned} \nu_x(\mathrm{d}A) &= Z_x^{-1} e^{-J_x(A)/\beta} b_x(\mathrm{d}A), \\ Z_x &= \mathbb{E}_{b_x} e^{-J_x(A)/\beta}. \end{aligned} \tag{2}$$

This distribution preserves the support of the learned prior but changes the relative weight of its plans. It does not assert that the cost reveals the original expert's density.

Proposition 4.1 (Noise-space planning target)**.** *Define*

$$p_x(z) = Z_x^{-1} \varphi(z) e^{-J_x(G_\psi(x,z))/\beta}. \tag{3}$$

Then $G_{\psi\#}p_x = \nu_x$. The normalizer in Equation (3) is the same as in Equation (2).

Proof. For any bounded measurable function f on chunks,

$$\begin{aligned} \mathbb{E}_{p_x} f(G_\psi(x,z)) &= Z_x^{-1} \mathbb{E}_\varphi\big[e^{-J_x(G_\psi(x,z))/\beta} f(G_\psi(x,z))\big] \\ &= Z_x^{-1} \mathbb{E}_{b_x}\big[e^{-J_x(A)/\beta} f(A)\big]. \end{aligned}$$

The second equality is the definition of a pushforward. □

The right-hand side of Equation (3) can be evaluated up to Z_x using a decoder call and a simulator rollout. No likelihood of b_x, inverse decoder, or Jacobian determinant is required.

There is also a path-space interpretation. Among laws Q on the decoder's revision paths with $Y_0 = C$, the minimizer of

$$\mathbb{E}_Q J_x(Y_1) + \beta\,\mathrm{KL}(Q\|B_x) \tag{4}$$

is the terminal tilt $\mathrm{d}Q^*/\mathrm{d}B_x = Z_x^{-1} e^{-J_x(Y_1)/\beta}$. Its terminal law is Equation (2). This is a soft-terminal stochastic-control problem, not the original SB problem with an expert-specified final marginal. The proof and an expected-cost comparison are in Section A.1.

4.2. What the dissipative prior contributes

The decoder can be pretrained by conditional DSBM from aligned old chunks to demonstrated future chunks. For a tractable physical prior, let $p_k = (q_k - q_{k-1})/\Delta t$ be the finite-difference velocity of a command sequence q_k. Set $q_{k+1} = a_k$ and initialize q_0, p_0 from the command history. A linear spring–damper model gives residuals

$$e_k(A) = p_{k+1} - p_k + \Delta t\,[K_h(q_k - m_h) + \Gamma_h p_k], \quad (5)$$

where K_h and Γ_h are fixed nonnegative stiffness and damping matrices and m_h is a reference location. An energy

$$U_h(A) = \tfrac{1}{2} \sum_k e_k(A)^\top Q_k^{-1} e_k(A) \quad (6)$$

uses positive-definite innovation covariances Q_k. With boundary commands fixed and any necessary quadratic anchoring, U_h is quadratic. The revision reference

$$\mathrm{d}Y_\tau = -D_h \nabla U_h(Y_\tau)\,\mathrm{d}\tau + \sqrt{2D_h}\,\mathrm{d}W_\tau, \quad (7)$$

with $D_h \succ 0$, is then affine Gaussian, so its conditioned bridges are tractable. W_τ is standard Brownian motion of chunk dimension. DSBM must use this reference's bridge samples and targets, not Brownian targets with the drift added only at deployment.

This construction translates assumptions about command sequences into a prior over edits. In the legacy Push-T interface, q_k is a target position followed by a PD controller (Action Bridge Policy Project, 2026). Dissipation in Equation (5) is therefore a command-space prior, not a certificate of physical contact passivity. Our sampler corrects against simulated outcomes precisely because the two are different. The main derivations below hold for any frozen G_ψ.

4.3. Learned proposals and a swap-symmetric correction

We temporarily suppress x. Let $q_\theta(z' \mid z)$ be a positive, evaluable proposal density, for example a diagonal Gaussian whose mean and variance depend on the current noise vector and context. Define the forward pair density

$$F_\theta(z, z') = p(z) q_\theta(z' \mid z), \qquad S(z, z') = (z', z). \quad (8)$$

The swap S is a volume-preserving involution. Its exact log-density ratio is

$$\begin{aligned} \ell_\theta(z, z') &= \log \frac{F_\theta(S(z, z'))}{F_\theta(z, z')} \\ &= \log \frac{\varphi(z')}{\varphi(z)} - \frac{J_x(G_\psi(x, z')) - J_x(G_\psi(x, z))}{\beta} \\ &\quad + \log \frac{q_\theta(z \mid z')}{q_\theta(z' \mid z)}. \end{aligned} \quad (9)$$

The exact acceptance probability $a_*(z, z') = \min\{1, e^{\ell_\theta(z, z')}\}$ yields an MH kernel that preserves p, for every proposal satisfying the usual measurability conditions.

For a fast variant, use an antisymmetric neural log-ratio

$$\begin{aligned} d_\phi(z, z') &= g_\phi(z, z') - g_\phi(z', z), \\ a_\phi(z, z') &= \min\{1, e^{d_\phi(z, z')}\}. \end{aligned} \quad (10)$$

The context is held fixed in both evaluations of g_ϕ. This parallels Ai-Sampler's use of symmetry, but the learned object is the density ratio of an explicit proposal pair. On acceptance, the new state is z'; on rejection, it is z. Denote the resulting approximate kernel by $T_{\theta,\phi}$.

During bootstrapping, exact simulator evaluations provide labels ℓ_θ. We fit

$$\mathcal{L}_{\mathrm{ratio}}(\phi; \theta) = \mathbb{E}_{(z, z') \sim F_\theta}[d_\phi(z, z') - \ell_\theta(z, z')]^2. \quad (11)$$

An adversarial alternative is binary pair classification with logit $-d_\phi$: forward pairs are positive, swapped pairs are negative. Its population optimum is also $d_\phi = \ell_\theta$, but a finite classifier loss does not by itself certify a small TV error. We use density-ratio regression when exact simulator labels are affordable.

4.4. From accepted probability flow to one-step error

Let $\bar{a} = \mathbb{E}_{F_\theta} a_\phi$ be the accepted probability mass and let

$$L_{\mathrm{bal}} = \mathbb{E}_{F_\theta}\big[a_\phi(d_\phi - \ell_\theta)\big].$$

The following results are stated for population expectations under the true fixed target p.

Proposition 4.2 (One-step balance bounds)**.** *If the stated expectations exist and $\bar{a} > 0$, then $L_{\mathrm{bal}} \geq 0$ and*

$$\mathrm{TV}(pT_{\theta,\phi}, p) \leq \sqrt{\bar{a} L_{\mathrm{bal}}/2}, \quad (12)$$

$$\mathrm{TV}(pT_{\theta,\phi}, p) \leq \mathbb{E}_{F_\theta}|a_\phi - a_*| \leq \sqrt{\mathcal{L}_{\mathrm{ratio}}}. \quad (13)$$

The proof in Section A.2 normalizes the accepted pair density $F_\theta a_\phi$. Its KL to its swap is $L_{\mathrm{bal}}/\bar{a}$. Importantly, the expansion retains *both* forward and reverse acceptance factors. We derive this expression independently rather than substituting the printed Ai-Sampler loss unchanged into a different kernel.

Why a balance loss is not a mixing objective. A kernel that almost always rejects changes little and can have a small one-step defect. Exact MH gives zero defect even for a proposal that barely moves. We therefore optimize proposal efficiency separately. With a fixed acceptance model, a bounded objective is

$$\mathcal{M}(\theta) = \mathbb{E}_{F_\theta}[a_\phi(z, z')\, m_x(z, z')], \tag{14}$$

where $0 \le m_x \le 1$ measures separation of decoded plans or their simulated behavior features. For example, it can be a clipped squared distance in fixed, normalized trajectory features. Maximizing accepted movement is a heuristic for exploration, not a proof of mixing. A likelihood-ratio gradient can optimize it without differentiating the simulator; see Section A.4.

4.5. An explicit safeguard against stationary bias

Let P_{ref} be exact independence MH with proposal φ. Apply this kernel with probability $\rho \in (0, 1]$ and the learned-acceptance kernel otherwise:

$$P = (1 - \rho)T_{\theta,\phi} + \rho P_{\mathrm{ref}}. \tag{15}$$

The occasional refresh includes a real simulator evaluation. Under our bounded-cost assumption, $P_{\mathrm{ref}}(z, \cdot) \ge e^{-J_{\max}/\beta} p(\cdot)$. Thus P contracts TV with coefficient at most $1 - \kappa$, where

$$\kappa = \rho e^{-J_{\max}/\beta} > 0. \tag{16}$$

Theorem 4.3 (Finite refinement and invariant-law error)**.** *Let $\epsilon_a = \mathbb{E}_{F_\theta}|a_\phi - a_*|$. The frozen kernel Equation (15) has a unique stationary law $\bar{p}$. For an arbitrary starting law λ and integer $n \ge 0$,*

$$\mathrm{TV}(\bar{p}, p) \le \frac{(1 - \rho)\epsilon_a}{\kappa}, \tag{17}$$

$$\mathrm{TV}(\lambda P^n, p) \le (1 - \kappa)^n\, \mathrm{TV}(\lambda, p) + \frac{(1 - \rho)\epsilon_a}{\kappa}[1 - (1 - \kappa)^n]. \tag{18}$$

The same bounds hold after decoding, by data processing.

The bounds are proved in Section A.3. They may be numerically weak when $J_{\max}/\beta$ is large. They formalize a trade-off, not a claim that the fast variant is automatically accurate. For an all-exact kernel, invariance is exact and only finite-step exploration remains to be evaluated.

4.6. Bootstrapping, deployment, and distillation

We alternate exact-sampling rounds, ratio fitting, and proposal improvement. All parameters defining the target, including ψ, are frozen during a round. Proposal adaptation between rounds is not treated as an already stationary adaptive chain. Finite chains provide approximate training samples; mixing diagnostics and independent validation contexts are required.

Algorithm 1 Ai-Plan training and receding-horizon use

Freeze pretrained decoder G_ψ and completion rule.
for each data-collection round **do**
  Execute learner to obtain actual contexts $x = (s, h, C, g)$.
  At each x, restore simulator copies for candidate costs.
  With proposal parameters frozen, collect exact-MH chains.
  Form proposal pairs and exact labels Equation (9).
  Fit d_ϕ using Equation (11); audit held-out ratios.
  Improve q_θ using Equation (14); then freeze it.
end for
At a new replan, initialize multiple noise states.
Run a budgeted exact or refreshed refinement chain.
Decode, validate, and execute a selected chunk prefix.
Retain its suffix; read the actual next observation.

An MCMC terminal sample approximates Equation (2). A best-feasible incumbent chosen from all evaluated candidates is an additional planning decision and is *not* a stationary sample. We report these modes separately. Newly found plans can be stored with the actual context and distilled into a conditional chunk reviser by supervised or bridge-matching updates. They are planner labels, not human ground truth. This distillation changes the decoder and is performed between frozen planning rounds.

Scope of the guarantees. The simulator defines the target; model error can therefore invalidate physical conclusions even when MH is exact. A fresh noisy rollout cost cannot be inserted into Equation (9) and assumed to yield exact MH for expected cost. Our primary setting is deterministic simulation; a fixed set of scenarios defines a different, well-specified deterministic target. Hard safety screening of the executed plan is separate from a finite soft penalty. Finally, an SB backward drift is not an evaluable reverse proposal density $q_\theta(z \mid z')$.

5. Experiments

Illustrative experimental section (not run). The following past-tense narrative and numbers describe a hypothetical successful study. They are not empirical evidence, benchmark scores, or significance estimates.

The illustrative study tested whether learning reduces simulator calls needed to find a useful correction, and whether approximate acceptance trades cost for a measurable stationary bias. All settings and numerical outcomes below are a proposed protocol and a constructed example of a successful

result.

5.1. Benchmarks and evaluation protocol

Analytic conditional planning distributions. We used two- and four-mode bounded energies on a compact two-dimensional diagnostic domain. Numerical integration supplied reference mode masses and discretized transition matrices. We varied proposal locality and the learned ratio error. This experiment separated high acceptance from genuine exploration: an identity-like proposal accepted almost every move but traversed modes slowly.

Ambiguous-route repair. A planar point system had two valid obstacle-avoidance routes. A plan was cached before a disturbance either left the route valid, displaced the goal, or blocked that route. We scored terminal goal error, constraint violations, and control variation through simulated execution. The tests included rare alternative routes, not just a symmetric two-mode case. Paired contexts shared a pre-fork history, allowing route proportions and cross-replan mode changes to be measured separately.

Contact-rich pushing. We used a planar Push-T-style task with absolute Cartesian pusher targets and a common PD tracking controller. A shifted object pose and an added fixture created recovery states after learner execution. A fixed evaluation cost used object-pose progress, completion, and finite penalties for excessive contact impulses. Each candidate was assessed from the same full snapshot by executing K commands and continuing for two more replans with a frozen policy. The recorded expert future was never reused as a corrective label after the learner changed the state.

We compared 50, 200, and 1,000 nominal demonstrations, five independent training seeds, and 200 held-out contexts per condition. For planning comparisons, every method received the same current observations, physical simulator, task cost, continuation horizon, and budget. We counted all simulated physical transitions, including rejected MH proposals, burn-in, refreshes, and lookahead. Reported candidate-equivalent budgets mean full candidate evaluations of the same horizon; wall-clock time was recorded separately.

5.2. Architecture and baselines

The illustrative DSBM prior used $H = 16$, $K = 8$, two observation/action-history frames, and a four-layer width-512 SiLU drift network with a 256-dimensional context embedding and a 64-dimensional revision-time embedding. Forward and reverse drifts were fitted for four outer matching rounds. Deployment used 16 stochastic revision steps, giving $d = 512$ Gaussian innovations for two-dimensional actions. A bounded action decoder mapped outputs into the command range. These are proposed settings, not properties of the older 52M-parameter actor in the project notes.

The learned noise proposal was a three-layer width-256 MLP predicting a diagonal-Gaussian mean residual and log-scale. A three-layer width-256 pair network produced the antisymmetric logit in Equation (10). Proposal scales were bounded; the acceptance classifier had no privileged future label. Both a Brownian decoder reference and the quadratic dissipative reference in Equation (7) were tested.

We compared unguided independent sampling, warm-started CEM, MPPI, pCN-MH in the *same decoder noise space*, a learned Gaussian proposal with exact MH, and learned acceptance with exact refresh. A geometry-matched FM decoder tested whether any gain was due to the prior rather than SB training. We also evaluated independent sampling with best-of-N selection, so MCMC was not compared only with a single sampled action.

Illustrative outcome. Table 1 shows how a successful paper could report an advantage at matched simulation cost. In this constructed example, the main improvement came from learning useful edits in the decoder's noise space. Moving from the geometry-matched FM decoder to DSBM gave a smaller increment than adding simulator-corrected refinement itself. This outcome would support a proposal-efficiency claim while limiting the claim that SB training is indispensable. The fast learned-acceptance row would additionally require a separate target-bias audit; recovery success is not a substitute for that audit.

5.3. Interpretation and ablations

In the constructed successful outcome, learned exact proposals improved recovery most when the old plan retained a useful contact or route. The fast version obtained a modest additional finite-budget improvement by rejecting poor proposals without always invoking the simulator; its mode-frequency error was higher. Raising the exact-refresh rate reduced that bias but consumed more of the budget. These are intended trade-offs, not guaranteed monotonic effects in arbitrary tasks.

Removing the movement objective produced high acceptance with slow mode exploration. Removing ratio calibration yielded attractive short-horizon plans but distorted target proportions. Replacing DSBM with a geometry-matched FM decoder retained much of the improvement: the useful result was the combination of a structured proposal and a correctness mechanism, not an exclusive advantage of SBs. Finally, a poorly chosen dissipative prior reduced recovery after large route invalidations, illustrating that damping should not be confused with the ability to abandon a stale decision.

Table 1: **Illustrative numbers only; no experiments run.** Constructed recovery success (%) with 200 demonstrations and equal-horizon simulator budgets. These values demonstrate reporting format, not benchmark measurements. No uncertainty estimates are implied.

Method	Route repair success		Pushing recovery success	
	Budget 32	Budget 128	Budget 32	Budget 128
Independent DSBM samples + best-of-N	61	77	48	66
Warm-started CEM	68	83	59	75
MPPI	70	85	61	78
pCN-MH, same DSBM decoder	73	87	64	80
Learned proposal + exact MH, FM decoder	79	90	71	86
Learned proposal + exact MH, DSBM decoder	82	92	75	89
Learned acceptance + exact refresh, DSBM decoder	85	93	79	90

Limitations of this experimental claim. A positive score at a fixed simulator budget would establish an efficient proposal for the chosen cost and simulator, not exact finite-time posterior sampling or real-robot safety. Acceptance rate, effective sample size, mode occupancy, and correction quality must be reported together. Failure of simple pCN or CEM to match the learned method is an empirical question; the hypothetical table is not evidence that such a failure occurs.

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A. Derivations and Assumptions

These appendices derive the stated population identities. They do not assume that the illustrative neural training procedure globally optimizes them. All kernels in a stationary-law argument are held fixed, and all target expectations refer to the same frozen context and decoder.

A.1. Terminal tilting and the Gibbs variational identity

Write $J = J_x(Y_1)$ and let Q^* satisfy $\mathrm{d}Q^*/\mathrm{d}B_x = e^{-J/\beta}/Z_x$. For any $Q \ll B_x$ with finite objective,

$$\beta\,\mathrm{KL}(Q\|Q^*) = \beta\mathbb{E}_Q \log \frac{\mathrm{d}Q}{\mathrm{d}B_x} + \mathbb{E}_Q J + \beta \log Z_x. \quad (19)$$

The left side is nonnegative, with equality exactly at $Q = Q^*$ up to null sets. This proves the minimizer of Equation (4). Since B_x is feasible,

$$\mathbb{E}_{Q^*} J + \beta\,\mathrm{KL}(Q^*\|B_x) \le \mathbb{E}_{B_x} J. \quad (20)$$

Hence $\mathbb{E}_{\nu_x} J_x \le \mathbb{E}_{b_x} J_x$. This is an ideal expected-cost comparison, not a finite-search or receding-horizon guarantee.

Conditioning fixes $Y_0 = C$. If sources are random and their marginal is constrained, normalization must be performed conditionally on each source; a single global cost tilt can otherwise change the source law. With a fixed source and any admissible terminal distribution, the endpoint-disintegration identity states that the minimum-KL path law uses the reference bridge between endpoints. This is the connection to SB theory described by Tang (2026), §4.1. It does not make the soft terminal objective identical to a two-fixed-marginal imitation problem.

A.2. Proof of the accepted-flow bound

For readability suppress θ, ϕ. Write $F(z, z') = p(z)q(z' \mid z)$, $F^S = F \circ S$, $a = a_\phi$, and $U = Fa$. The acceptance mass is $\bar{a} = \int U$. The density of $pT - p$ is

$$(pT - p)(z) = \int [U(z', z) - U(z, z')]\,\mathrm{d}z'. \quad (21)$$

Taking absolute values, integrating, and dividing by two gives

$$\mathrm{TV}(pT, p) \le \tfrac{1}{2} \int |U - U^S| = \bar{a}\,\mathrm{TV}(U/\bar{a}, U^S/\bar{a}). \quad (22)$$

The two pair measures have equal mass because the swap has unit Jacobian. Antisymmetry of d gives

$$\log a(z, z') - \log a(z', z) = \min(0, d) - \min(0, -d) = d. \quad (23)$$

Since $\ell = \log(F^S/F)$, it follows that

$$\mathrm{KL}(U/\bar{a}\|U^S/\bar{a}) = \bar{a}^{-1}\mathbb{E}_F[a(d - \ell)]. \quad (24)$$

The expectation on the right is nonnegative as a consequence, even though an individual summand or finite-sample estimate can be negative. Pinsker's inequality in our TV convention yields Equation (12).

For the second bound, exact MH satisfies $pT_* = p$. For any event E,

$$(pT - pT_*)(E) = \mathbb{E}_F[(a - a_*)\{\mathbf{1}_E(z') - \mathbf{1}_E(z)\}]. \quad (25)$$

The braces have magnitude at most one, so $\mathrm{TV}(pT, p) \le \mathbb{E}_F|a - a_*|$. The function $t \mapsto \min(1, e^t)$ is globally 1-Lipschitz. Therefore

$$\mathbb{E}_F|a - a_*| \le \mathbb{E}_F|d - \ell| \le \sqrt{\mathbb{E}_F(d - \ell)^2}, \quad (26)$$

proving Equation (13).

Relation to the original Ai-Sampler objective. The transferable structure is a discrepancy between probability flow and its involution, followed by a one-step invariance bound. In the accepted-density derivation above, the log ratio contains $\log a(z, z') - \log a(z', z)$ and a normalization by $\bar{a}$. We retain these terms explicitly. We do not claim that this objective is literally Equation (23) of Egorov et al. (2024), nor that its original discriminator parameterization applies unchanged to our stochastic proposal.

A.3. Exact refresh, contraction, and finite-step bias

Let $w(z) = e^{-J_x(G_\psi(x,z))/\beta} \in [e^{-J_{\max}/\beta}, 1]$. The continuous part of the independence-MH transition is

$$\varphi(z') \min\{1, w(z')/w(z)\}\,\mathrm{d}z'. \quad (27)$$

Because $w(z) \le 1$, this is at least $\varphi(z')w(z')\,\mathrm{d}z' = Z_x p(z')\,\mathrm{d}z'$. Consequently,

$$P_{\mathrm{ref}}(z, \cdot) \ge Z_x p(\cdot) \ge e^{-J_{\max}/\beta} p(\cdot). \quad (28)$$

The mixture P therefore satisfies $P(z, \cdot) \ge \kappa p(\cdot)$. Write $P = \kappa p + (1 - \kappa)\widetilde{P}$, where $\widetilde{P}$ is another Markov kernel. For probability laws u, v,

$$\mathrm{TV}(uP, vP) \le (1 - \kappa)\,\mathrm{TV}(u, v). \quad (29)$$

Uniform minorization yields a unique invariant probability law. Since $pP_{\mathrm{ref}} = p$,

$$\mathrm{TV}(pP, p) \le (1 - \rho)\epsilon_a. \quad (30)$$

For $e_n = \mathrm{TV}(\lambda P^n, p)$, the triangle inequality and Equation (29) give

$$e_{n+1} \le (1 - \kappa)e_n + (1 - \rho)\epsilon_a. \quad (31)$$

Solving this scalar recursion proves Equation (18); its limit proves Equation (17). Applying the deterministic decoder cannot increase TV.

Why the mixing denominator is necessary. For any incorrect law q, the kernel $P_\varepsilon(z,\cdot) = (1-\varepsilon)\delta_z + \varepsilon q$ has stationary law q. Yet $\mathrm{TV}(pP_\varepsilon, p) = \varepsilon\,\mathrm{TV}(p,q)$ can approach zero. This example prevents interpreting an unscaled one-step loss as a stationary certificate. Our explicit denominator plays the same logical role as the minorization term in Neklyudov et al. (2019).

A.4. Black-box proposal training and target sampling

With ϕ frozen, p fixed, and $v_x(z,z') = a_\phi(z,z')m_x(z,z')$, a score-function derivative is

$$\nabla_\theta \mathcal{M} = \mathbb{E}_{p(z)q_\theta(z'|z)}[v_x(z,z')\nabla_\theta \log q_\theta(z' \mid z)]. \quad (32)$$

Any baseline depending on (x,z) but not on z' can be subtracted from v_x. The sample z' is held fixed inside the log-density derivative. This includes the effect of the sampling distribution without requiring a derivative of simulated outcomes. In contrast, differentiating an exact MH acceptance that depends explicitly on q_θ would require additional terms; the displayed update intentionally uses the frozen learned acceptance during proposal improvement.

The ratio target Equation (9) is recomputed after proposal changes. Ratio fitting uses simulator-cost *values*, not gradients. Exact-MH bootstrap chains must be long enough or diversified enough to approximate the target. Their samples are correlated; finite-chain bias and neural function-class limitations are not controlled by Theorem 4.3. Evaluate ratio error on independent contexts and exact-correction chains, not only on the proposal's own easiest examples.

A.5. pCN as a non-learned exact baseline

For step parameter $0 < \eta \le 1$, pCN proposes

$$z' = \sqrt{1-\eta^2}\,z + \eta\epsilon, \qquad \epsilon \sim \mathcal{N}(0, I_d). \quad (33)$$

The Gaussian prior satisfies $\varphi(z)q(z' \mid z) = \varphi(z')q(z \mid z')$, so the MH ratio reduces to $\exp[-(J_x(G_\psi(x,z')) - J_x(G_\psi(x,z)))/\beta]$. This makes it a strong baseline: learning must improve the geometry of exploration rather than merely rediscover a correct accept/reject rule (Cotter et al., 2013).

B. Reproducibility Specification and Boundaries

Three distinct iteration axes. Physical execution advances the robot; SB revision transforms a complete chunk; MCMC refinement changes the random input used by the frozen SB decoder. The index n in Equation (18) counts MCMC steps, not robot commands or DSBM training rounds.

Frozen versus learned objects. Within an exactness argument, freeze the scene x, simulator, cost, decoder G_ψ, and proposal/acceptance snapshot. The initial noise vector may be sampled from φ or warm-started; it need not be distributed as p_x. This is why a finite-time term appears in Equation (18). Starting at the next robot observation defines a new target and a new chain, not continued stationary sampling from the old target.

Action bounds and feasibility. Use a fixed smooth bounded decoder or a fixed projection before sending targets to the environment, and include that map in G_ψ when defining the cost. The pushforward proof remains valid. Clipping can make the chunk law singular, but it does not prevent evaluating the noise-space target. Final safety checks may reject or replace a planned execution; the resulting controller is then not identical to a posterior sample from Equation (2).

Budget accounting. One candidate evaluation includes decoder execution and the specified physical continuation. Rejected proposals and exact refreshes count. Training-time simulator calls and deployment-time calls are reported separately. A learned acceptance model can appear efficient only if its amortization cost and failure modes are visible. The exact-MH baseline should use the *same* learned proposal to separate proposal learning from acceptance approximation.

Statistical reporting. Replace the constructed table with per-seed measurements before any submission. Report success, final cost, simulated constraint violations, wall-clock time, target-mode discrepancy on tractable diagnostics, and autocorrelation/ESS of relevant observables. High acceptance is not evidence of exploration; best-found cost is not evidence of stationary correctness.

What would constitute a contribution. The pushforward identity, Gibbs tilt, MH correction, and minorization argument are established mathematical tools. The proposed contribution is a conditional, learned refinement mechanism in a behavior-cloned generator's noise space, together with a measurable calibration–mixing–simulation-budget tradeoff. It remains useful if a matched FM decoder performs similarly; that outcome would limit the SB-specific claim rather than invalidate the sampler.